

Engineering Curves: Construction of Epicycloid

Rolling Circle Method with Tangent & Normal

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Problem Statement

Problem

A generating circle of diameter 30 mm rolls externally without slipping on a directing circle of diameter 90 mm. Construct one complete epicycloid arch corresponding to $\angle AOB = 120^\circ$, divide the arch into 12 equal angular parts, and retain every construction element in all subsequent views.

Given Data

Diameter of directing circle = 90 mm, $R = 45$ mm,
Diameter of generating circle = 30 mm, $r = 15$ mm,
 $\angle AOB = 120^\circ$, Number of divisions = 12.

$$OC_0 = R + r = 45 + 15 = 60 \text{ mm},$$

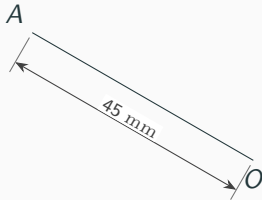
$$AC_0 = r = 15 \text{ mm},$$

$$\Delta\theta = \frac{120^\circ}{12} = 10^\circ,$$

$$\text{Generating-circle division angle} = \frac{360^\circ}{12} = 30^\circ.$$

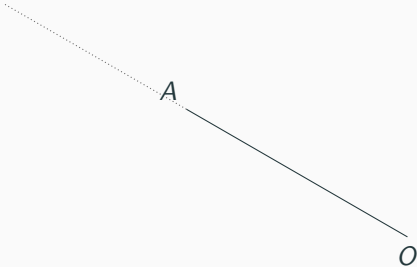
Step 1: Draw OA

$OA = 45 \text{ mm.}$



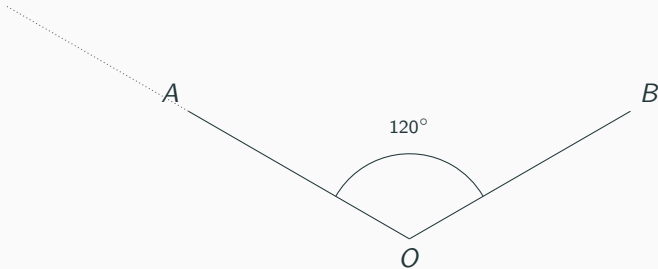
Step 2: Extend OA

Extend OA .



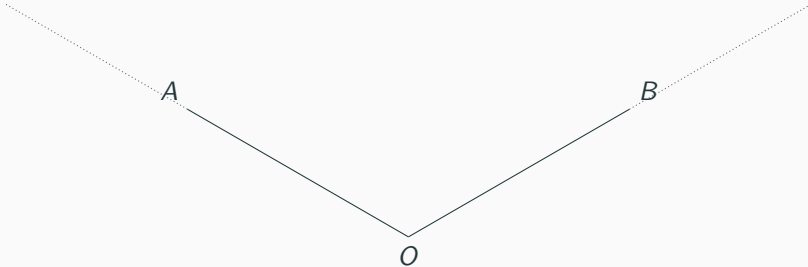
Step 3: Draw OB at 120°

$$\angle AOB = 120^\circ.$$



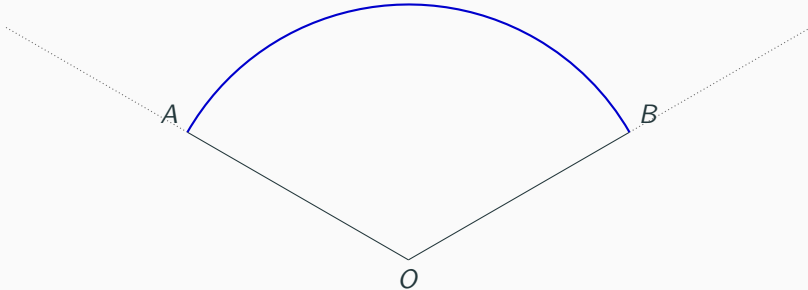
Step 4: Extend OB

Extend OB .



Step 5: Draw the directing arc AB

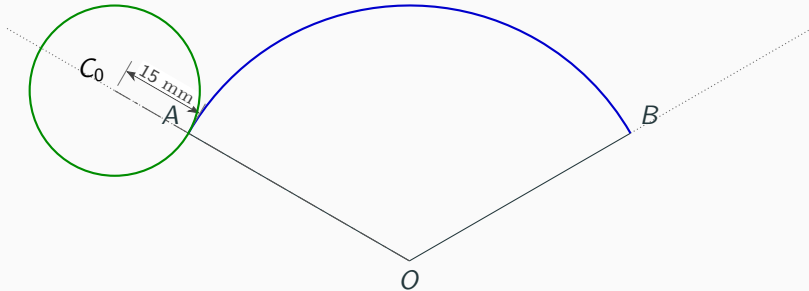
Blue arc AB : directing circle.



Step 6: Locate C_0 and the generating circle

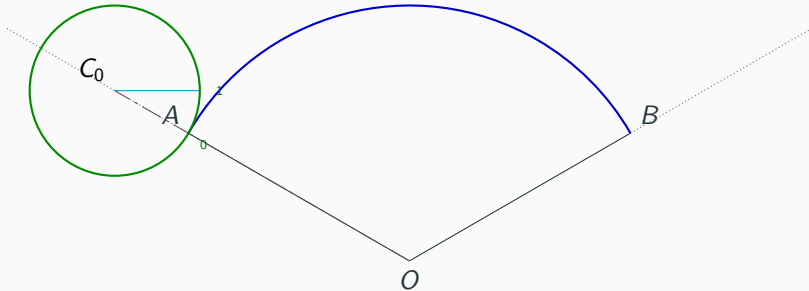
$AC_0 = 15 \text{ mm.}$

Green: generating circle.



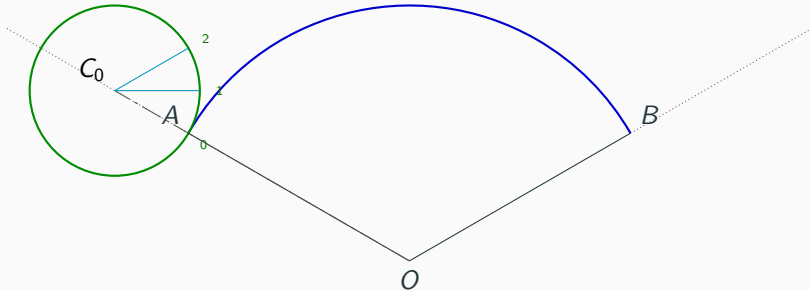
Step 7: Divide the generating circle into 12 equal parts

Green: generating circle.
30° divisions.



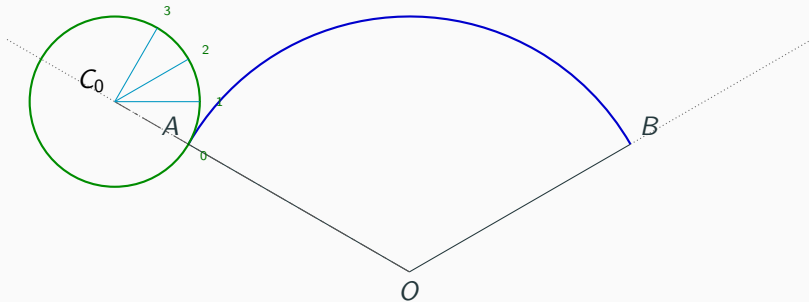
Step 7: Divide the generating circle into 12 equal parts

Green: generating circle.
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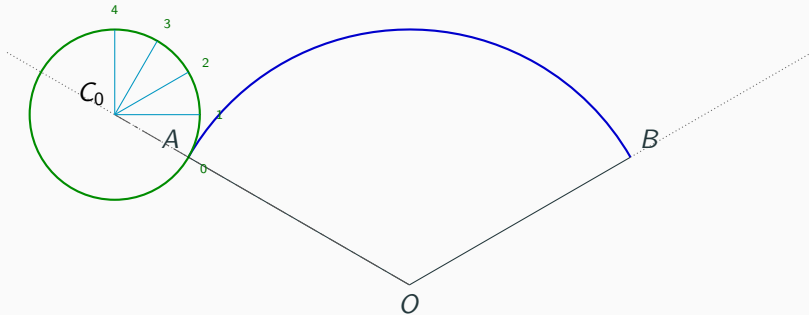
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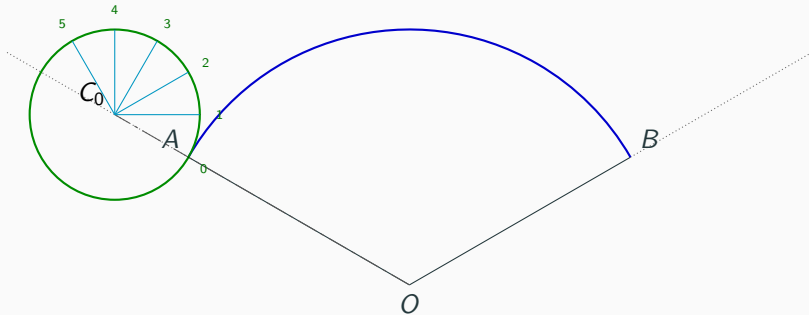
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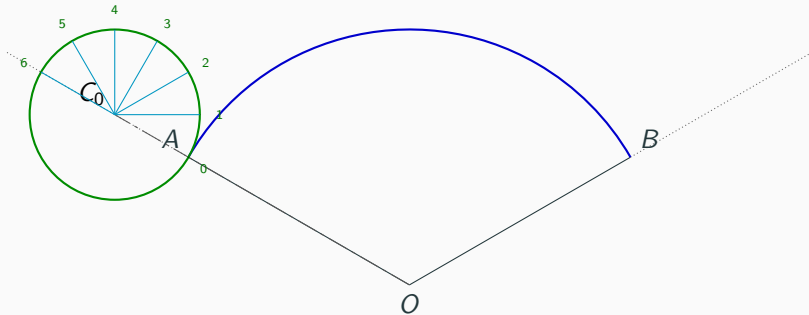
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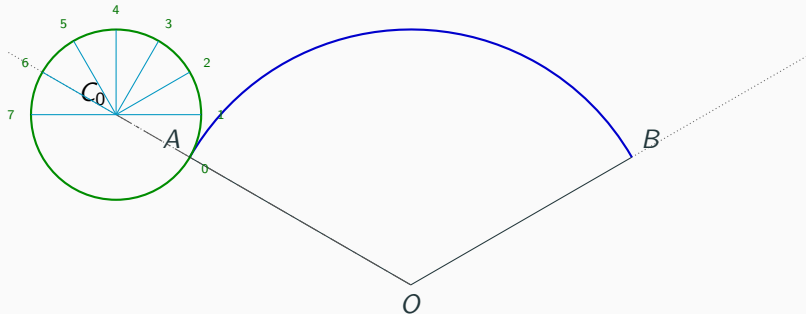
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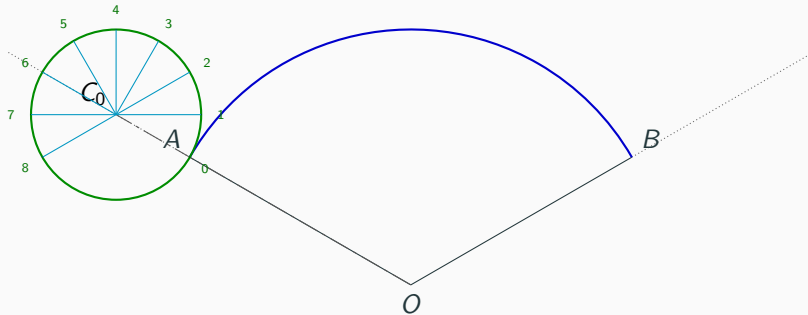
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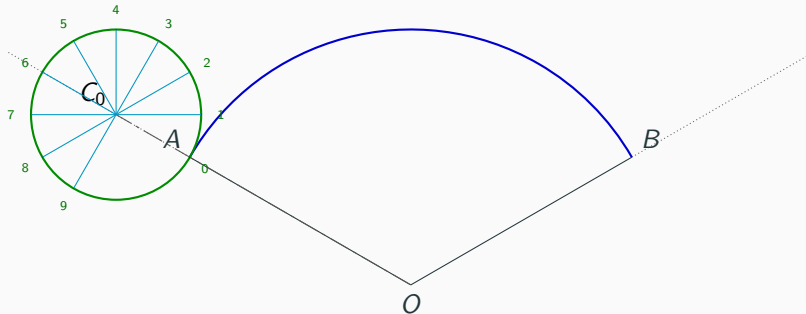
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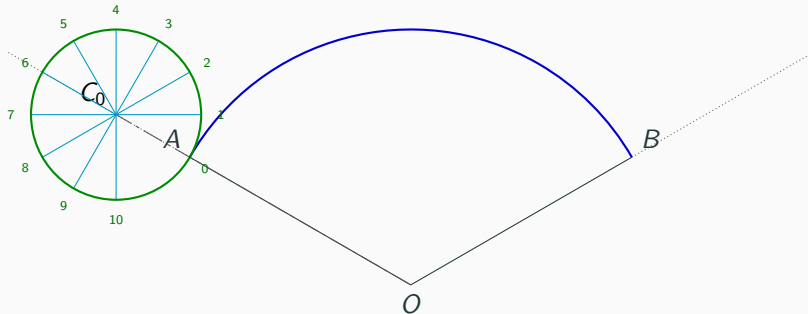
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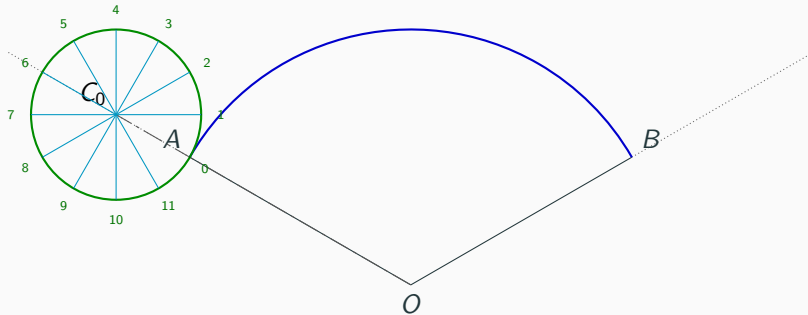
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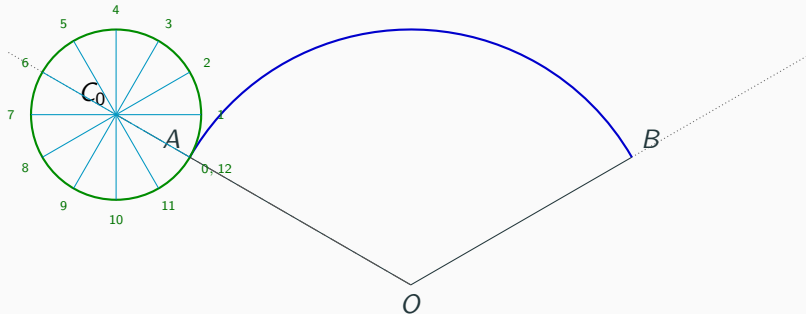
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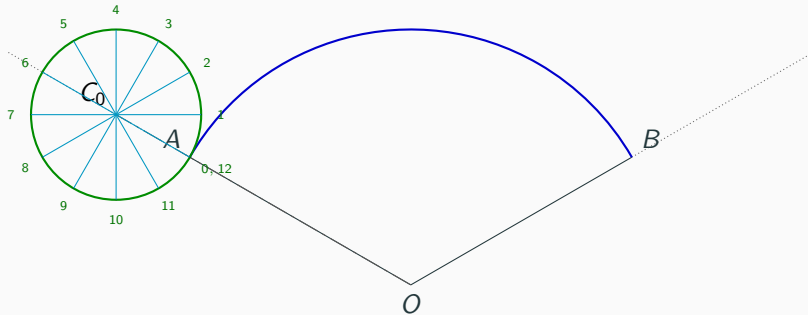
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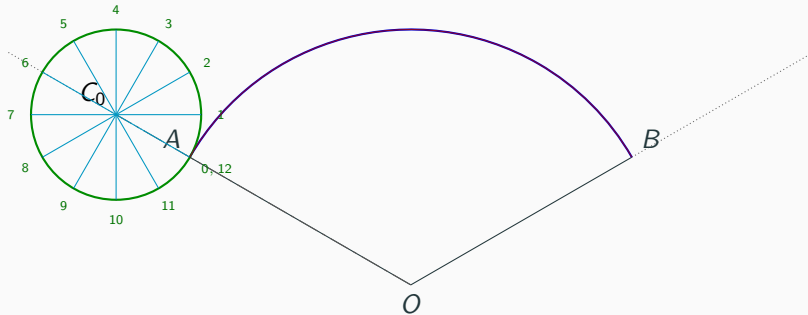
Step 8: Complete the generating-circle divisions

Generating-circle divisions.



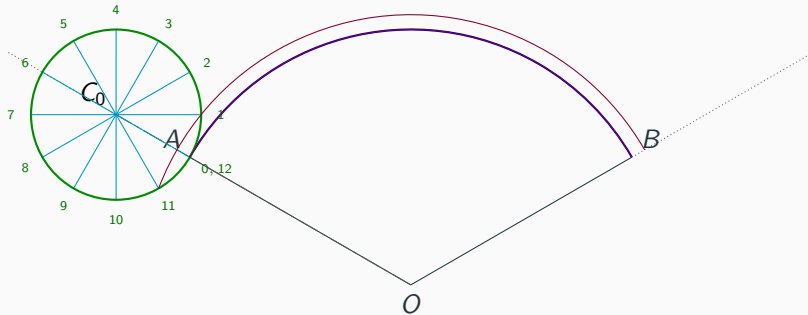
Step 9: Draw concentric construction arcs

Concentric construction arc.



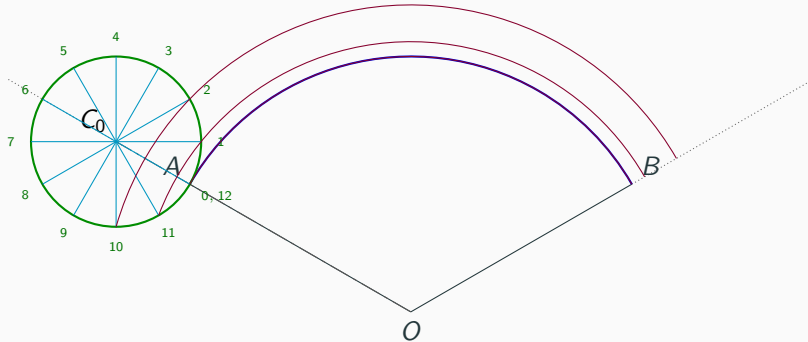
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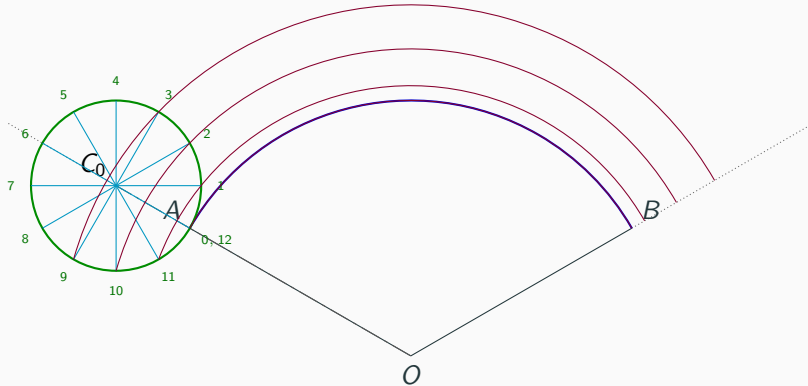
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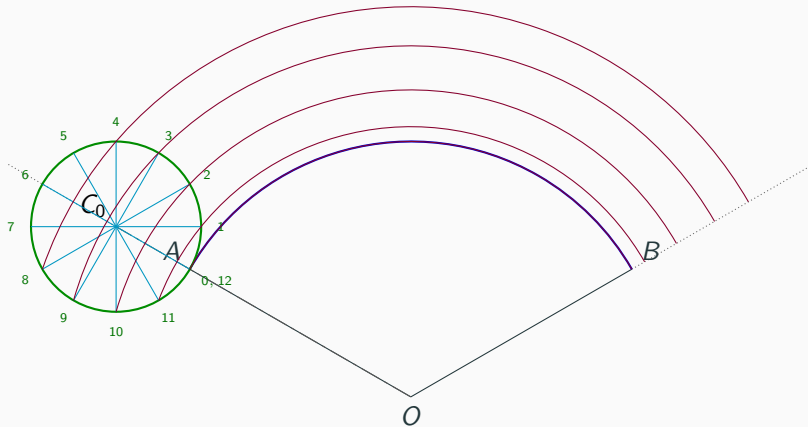
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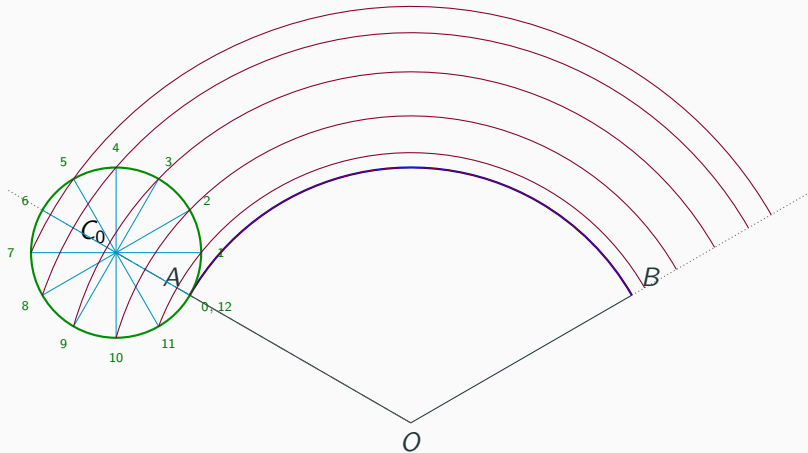
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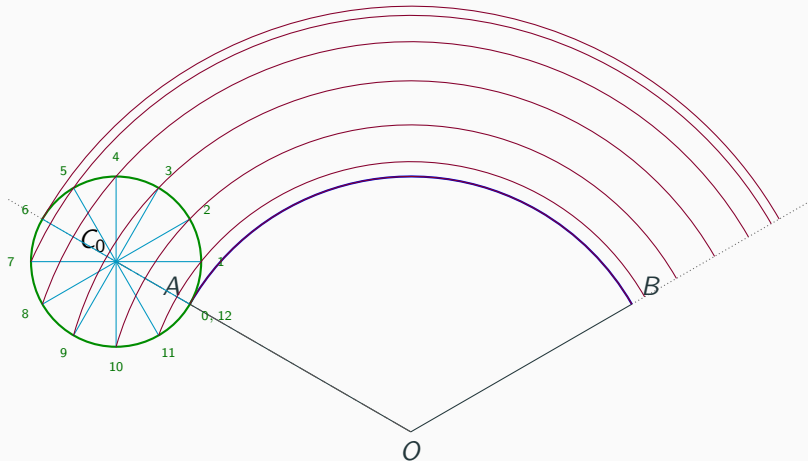
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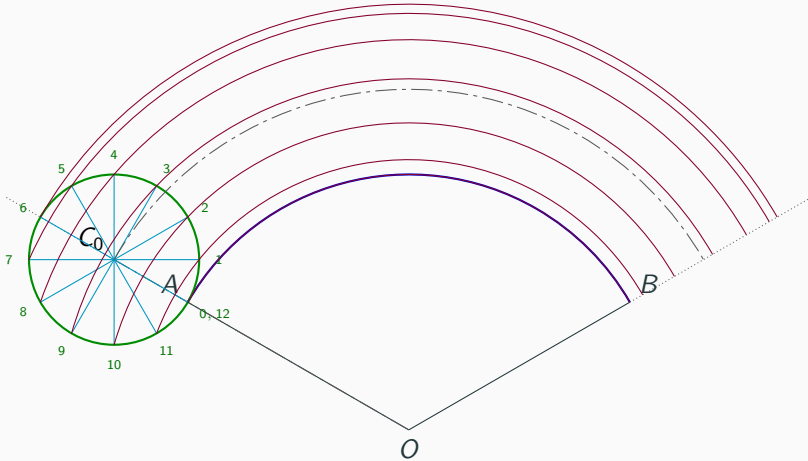
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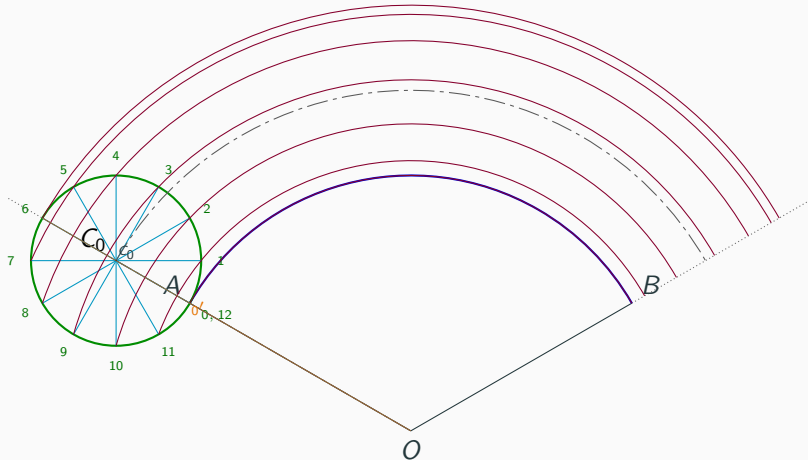
Step 10: Draw the centre arc through C_0

Centre curve: ISO axis line.



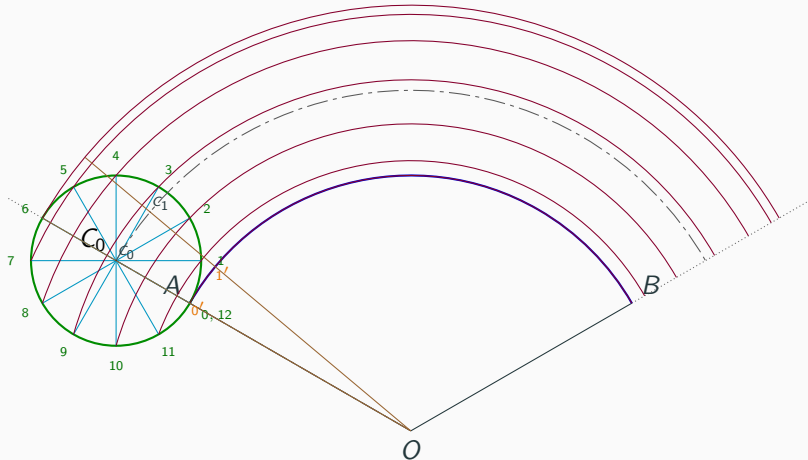
Step 11: Divide $\angle AOB$ into 12 equal parts

10° divisions.



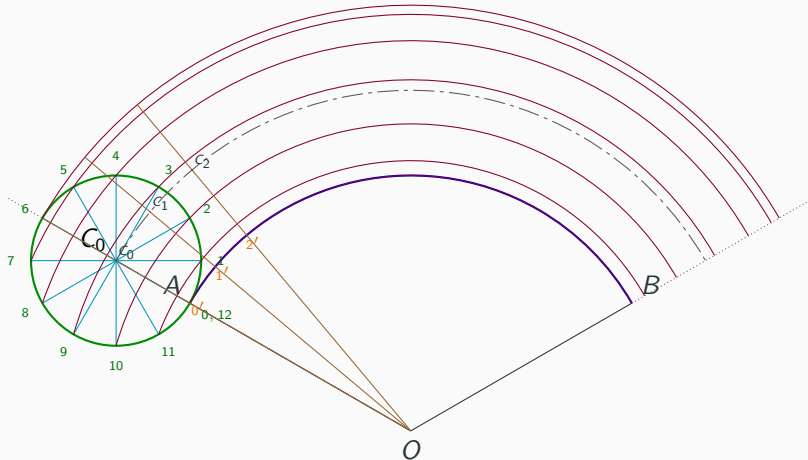
Step 11: Divide $\angle AOB$ into 12 equal parts

10^0 divisions.



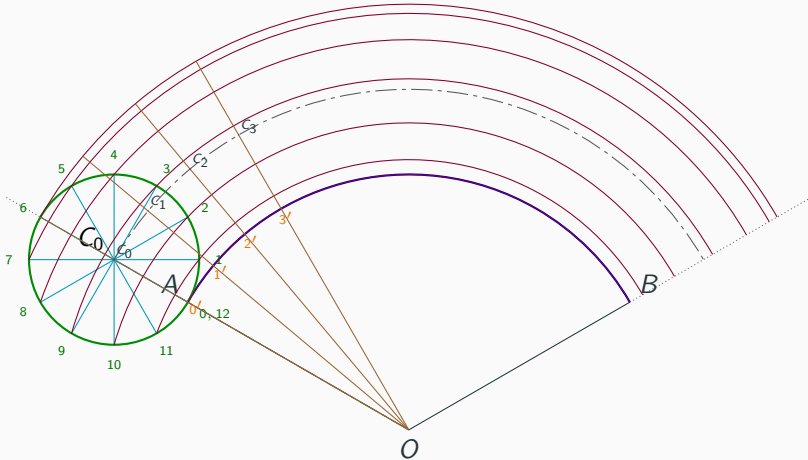
Step 11: Divide $\angle AOB$ into 12 equal parts

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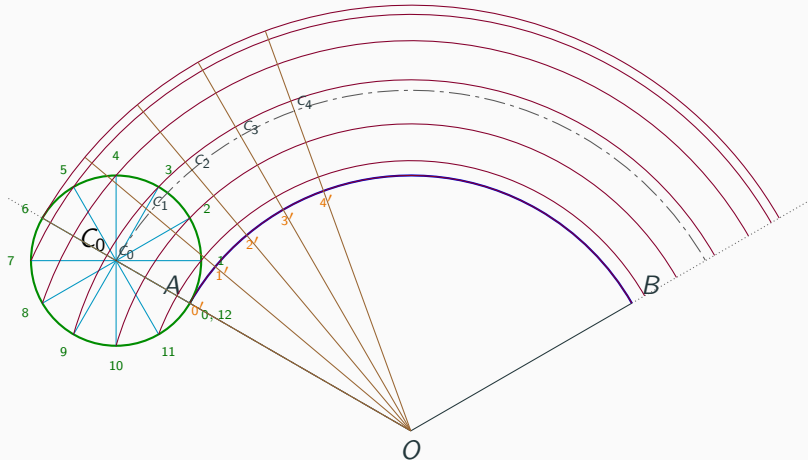
Step 11: Divide $\angle AOB$ into 12 equal parts

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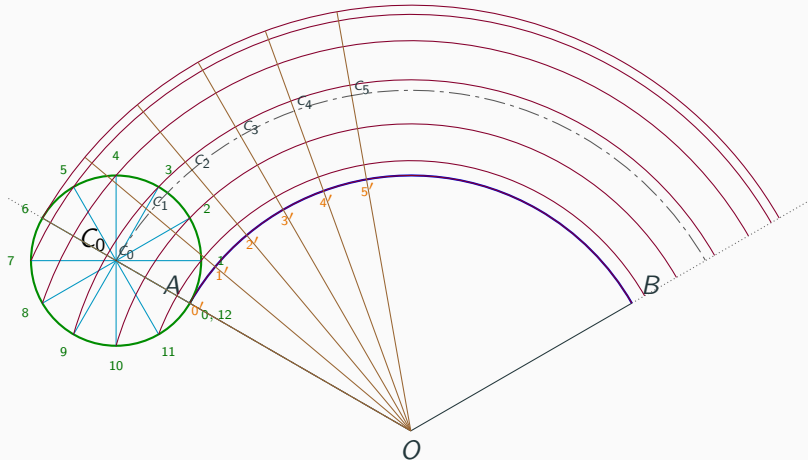
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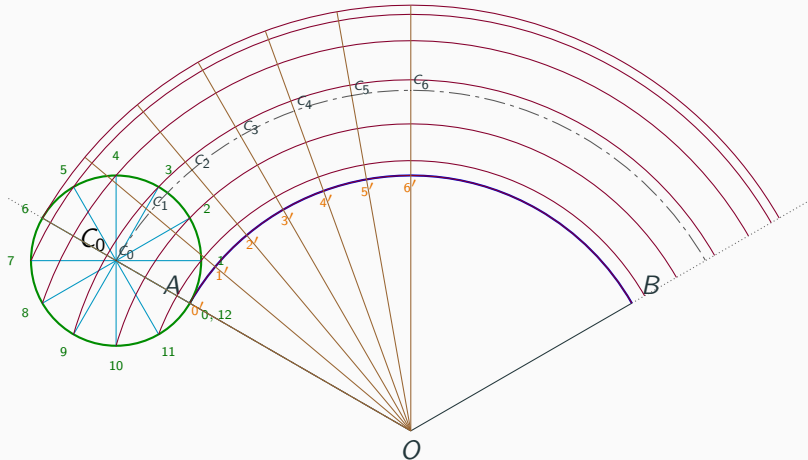
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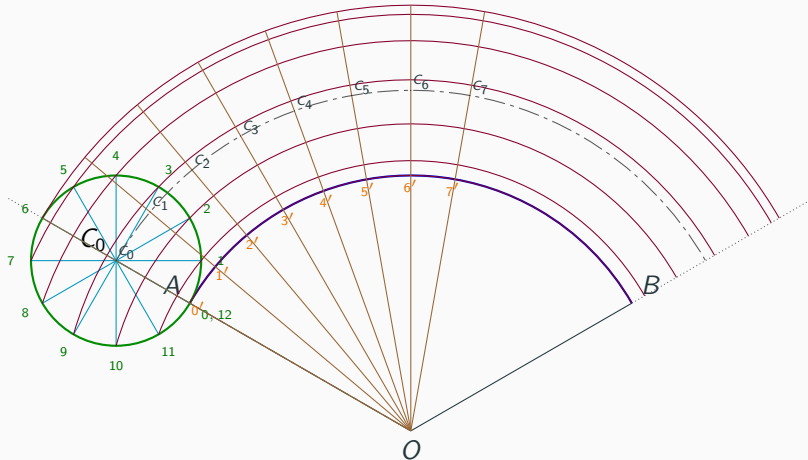
Step 11: Divide $\angle AOB$ into 12 equal parts

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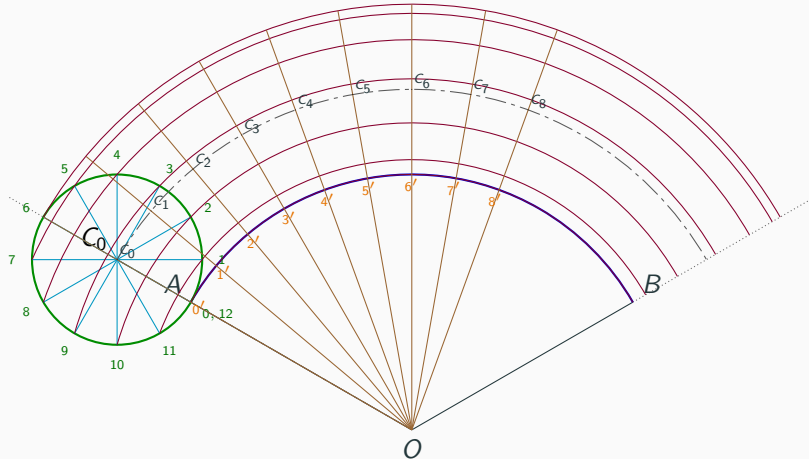
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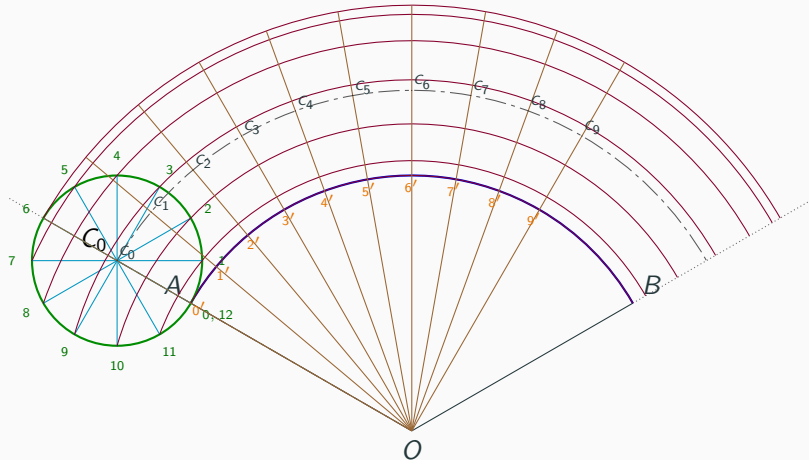
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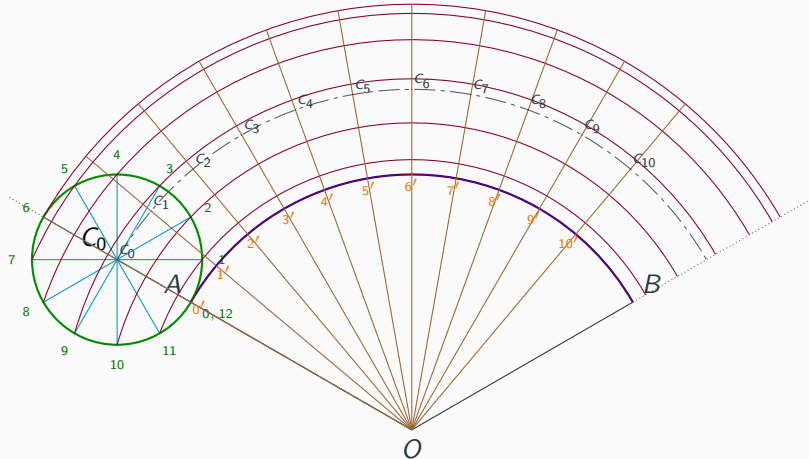
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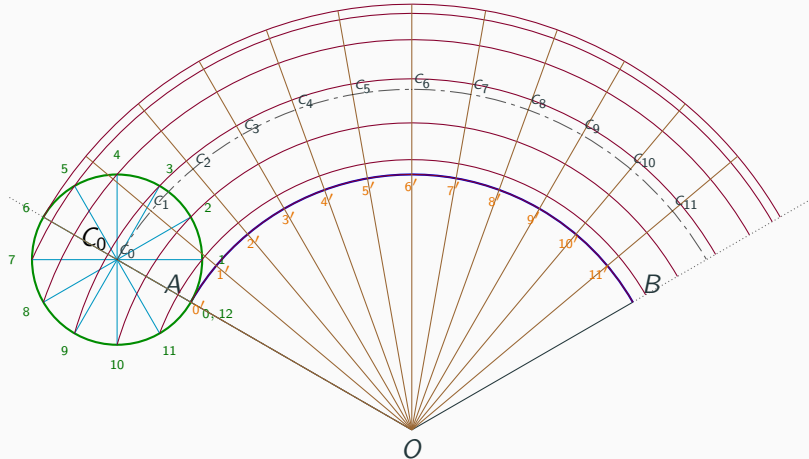
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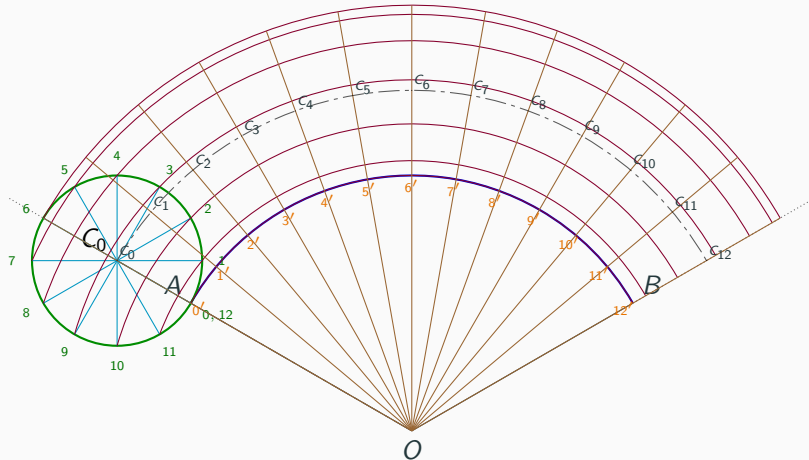
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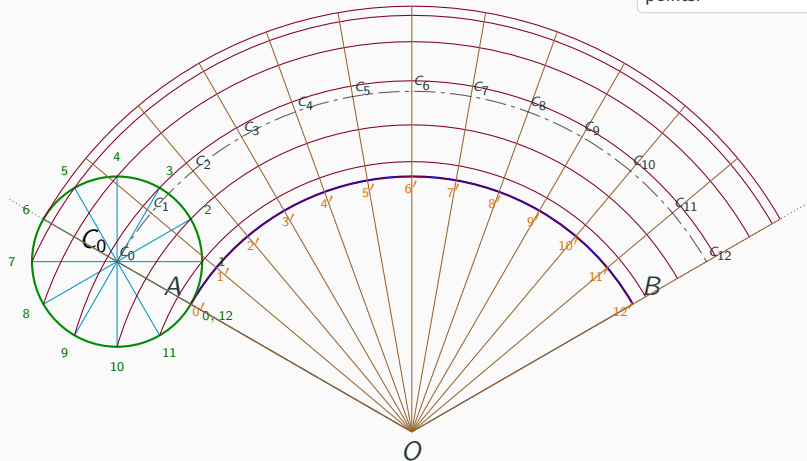


Step 11: Divide $\angle AOB$ into 12 equal parts

10° divisions.



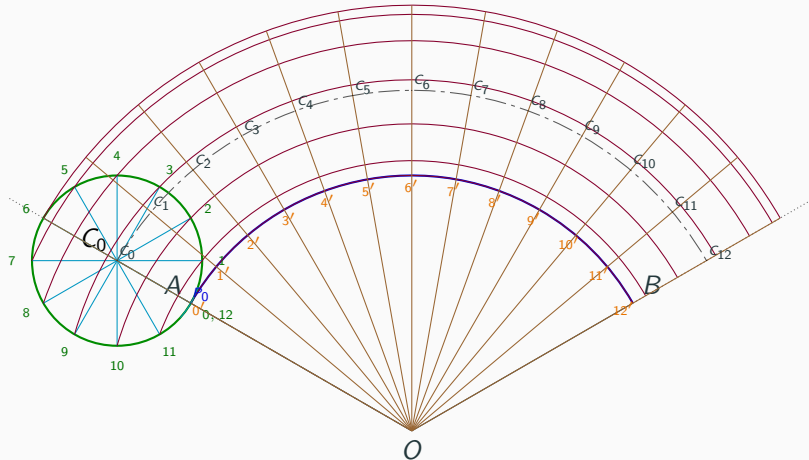
Step 12: Complete the angular divisions and points



Division points and centre points.

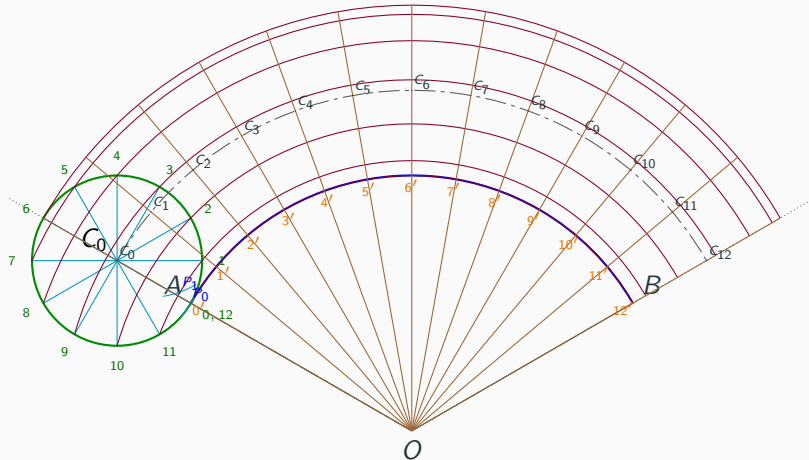
Step 13: Draw the small cutting arc and locate P_i

Cyan arc: radius 15 mm.



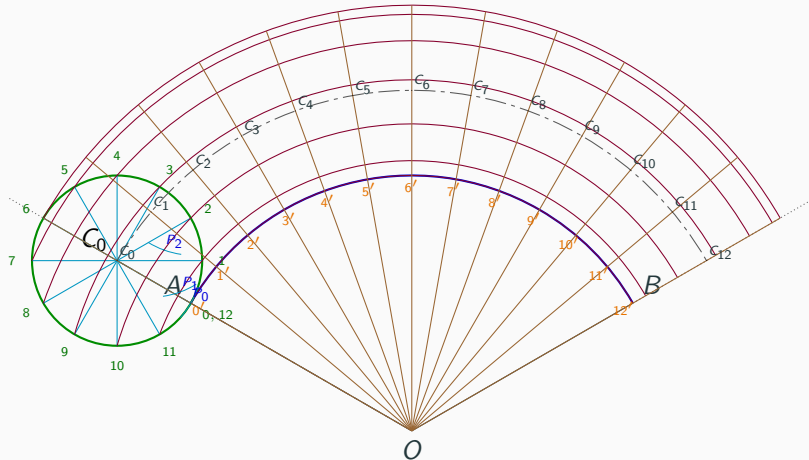
Step 13: Draw the small cutting arc and locate P_i

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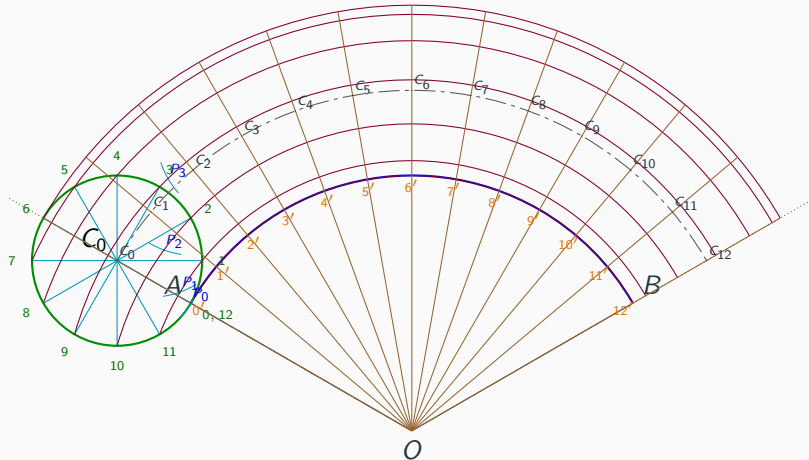
Step 13: Draw the small cutting arc and locate P_i

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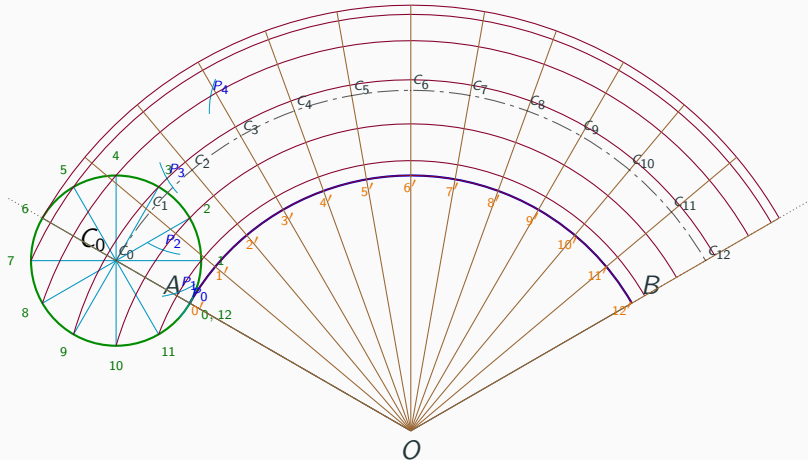
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Cyan arc: radius 15 mm.



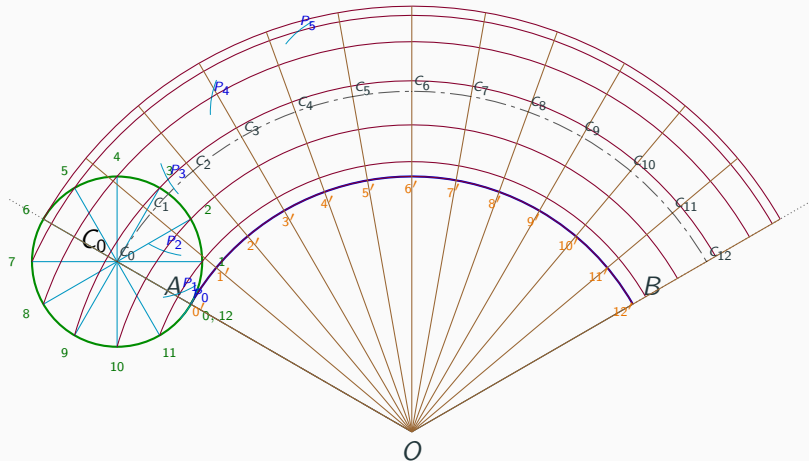
Step 13: Draw the small cutting arc and locate P_i

Cyan arc: radius 15 mm.

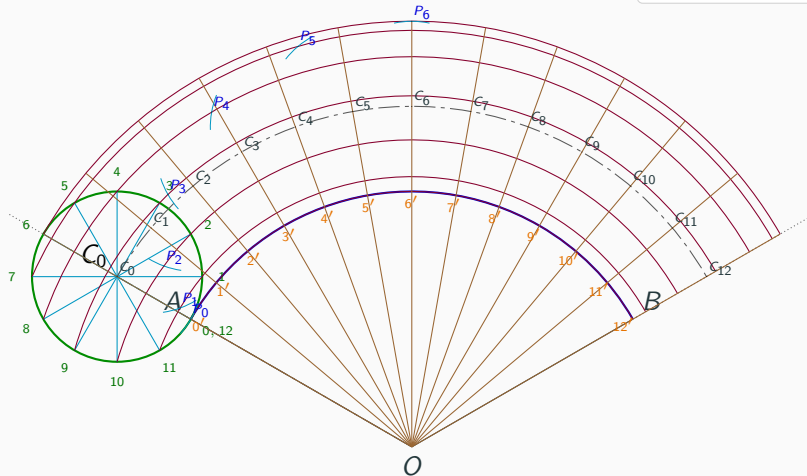


Step 13: Draw the small cutting arc and locate P_i

Cyan arc: radius 15 mm.

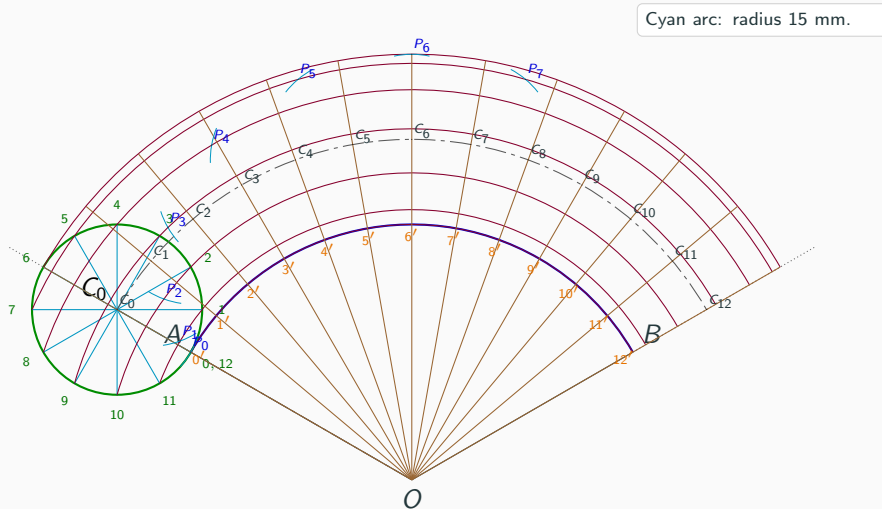


Step 13: Draw the small cutting arc and locate P_i



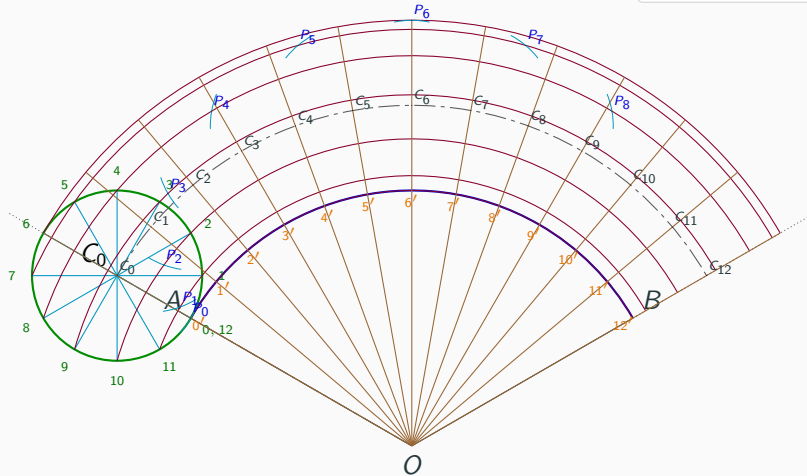
Cyan arc: radius 15 mm.

Step 13: Draw the small cutting arc and locate P_i

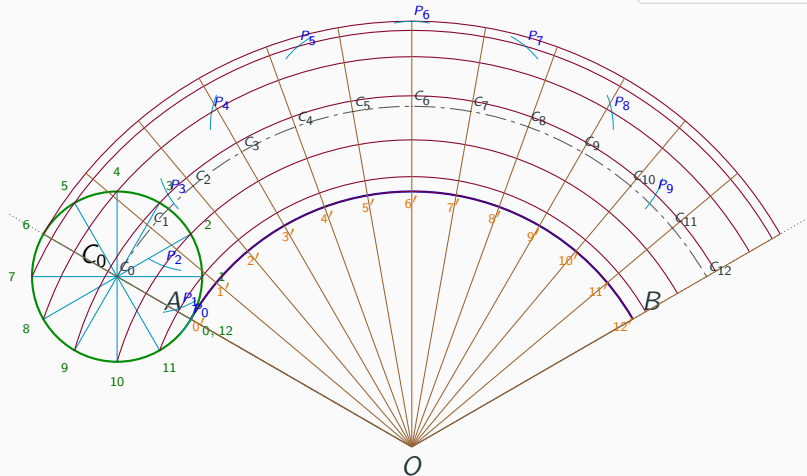


Step 13: Draw the small cutting arc and locate P_i

Cyan arc: radius 15 mm.

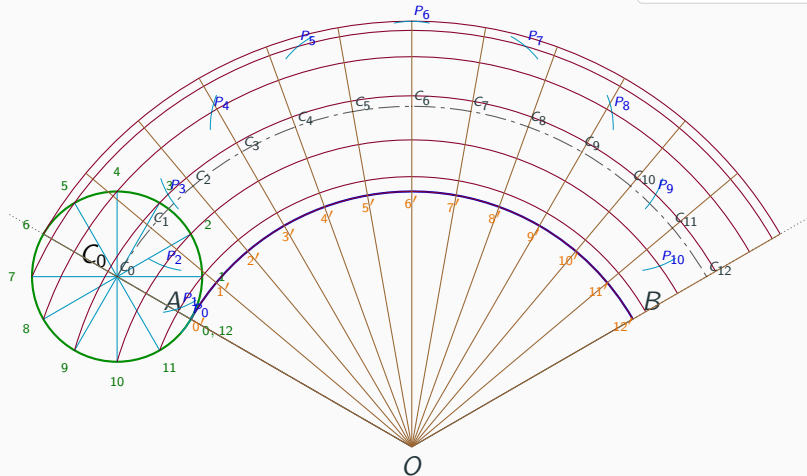


Step 13: Draw the small cutting arc and locate P_i



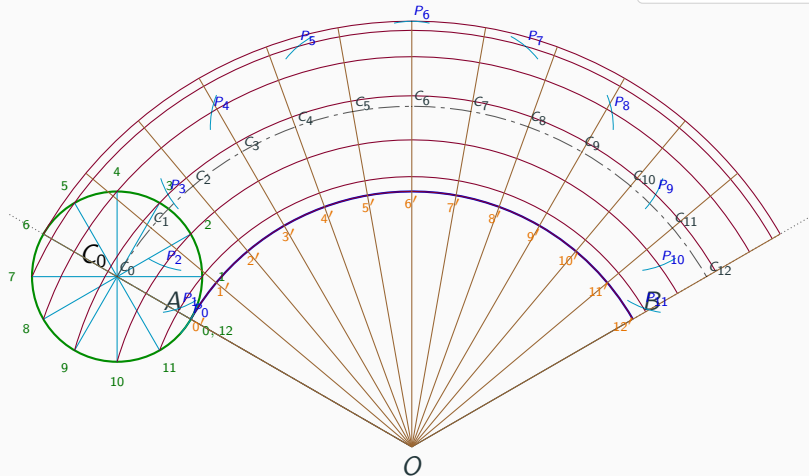
Cyan arc: radius 15 mm.

Step 13: Draw the small cutting arc and locate P_i



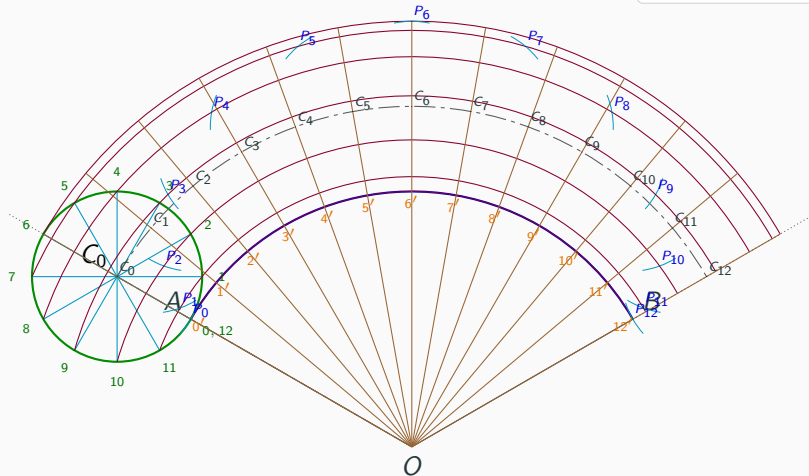
Cyan arc: radius 15 mm.

Step 13: Draw the small cutting arc and locate P_i



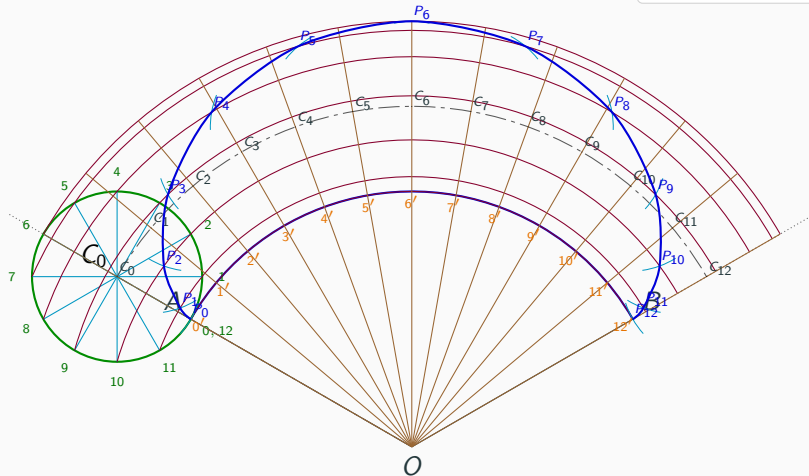
Cyan arc: radius 15 mm.

Step 13: Draw the small cutting arc and locate P_i



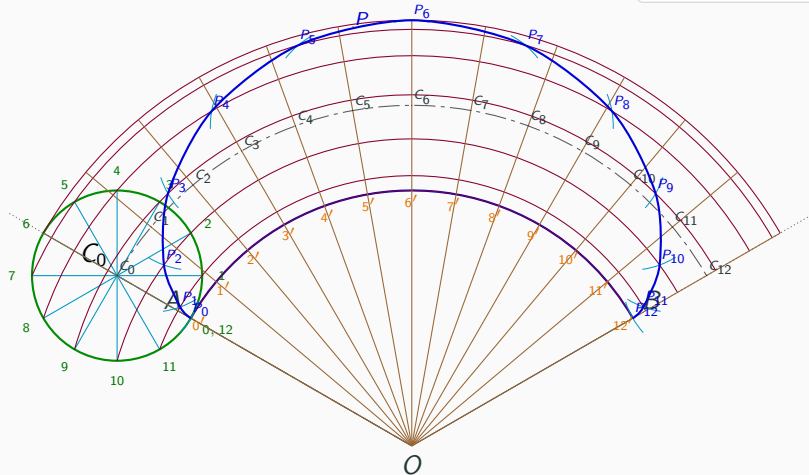
Cyan arc: radius 15 mm.

Step 14: Draw the epicycloid



Blue curve: epicycloid.

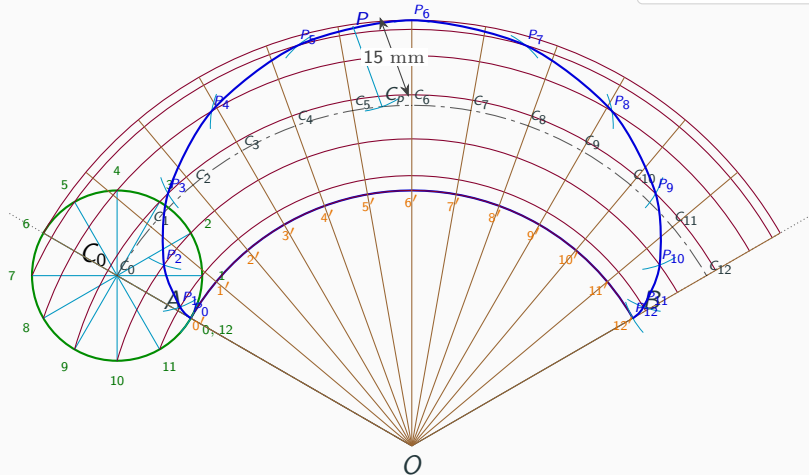
Step 15: Select a point P on the epicycloid



Selected point P .

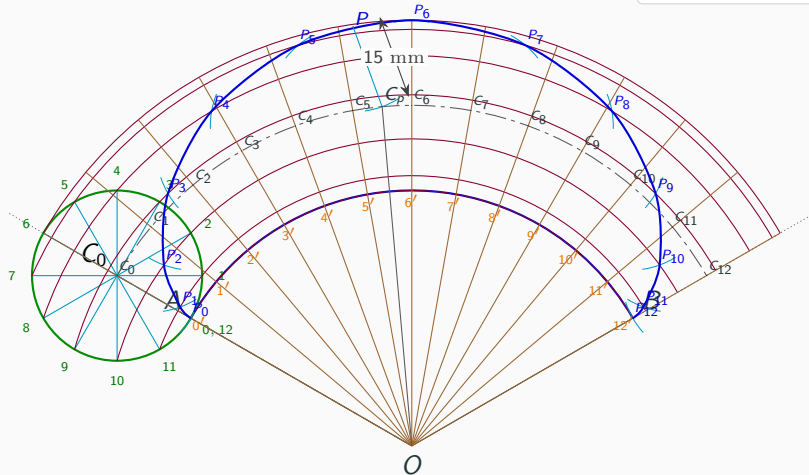
Step 16: Draw the 15 mm arc from P to the centre curve

$$PC_P = 15 \text{ mm}$$



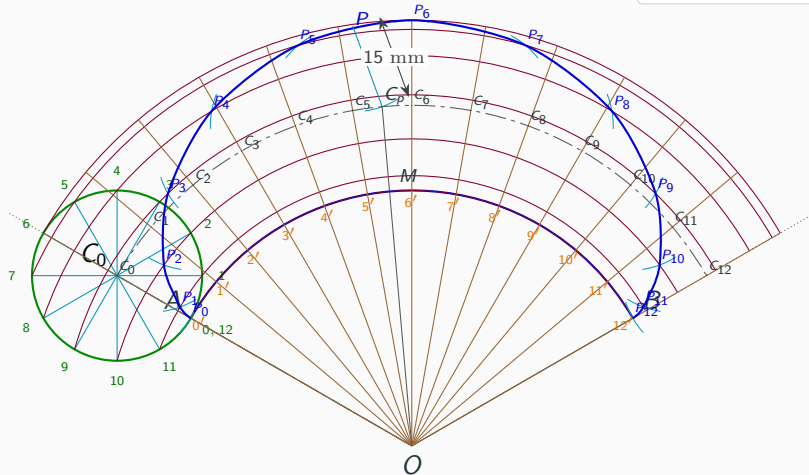
Step 17: Join C_p to O

Join C_p to O .

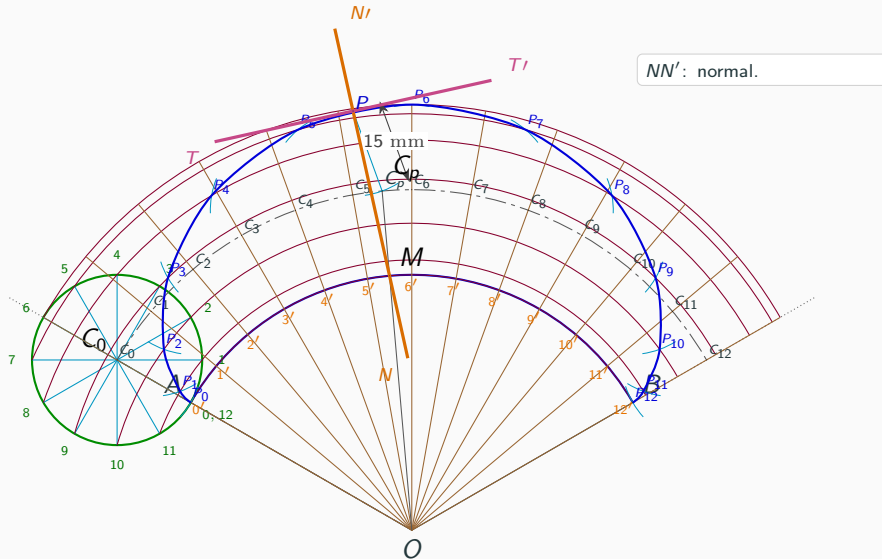


Step 18: Mark M on the directing circle

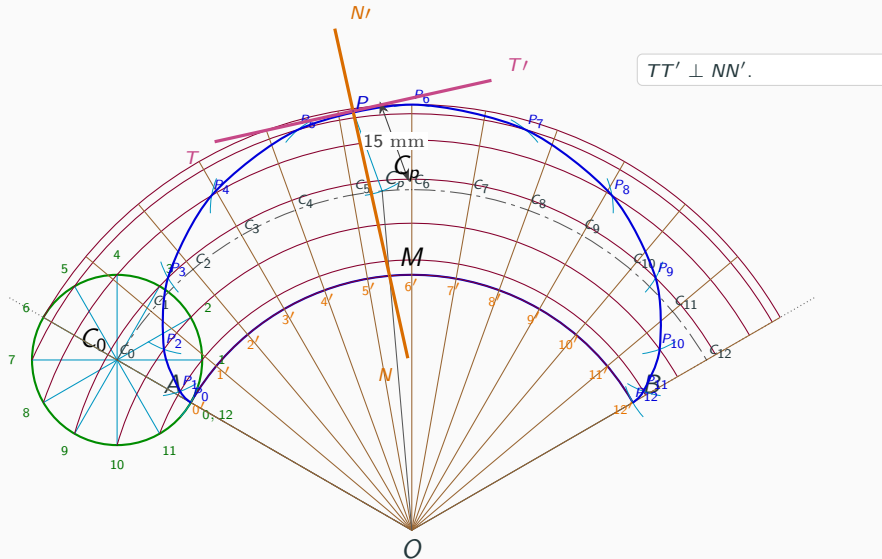
$$M = OC_p \cap AB.$$



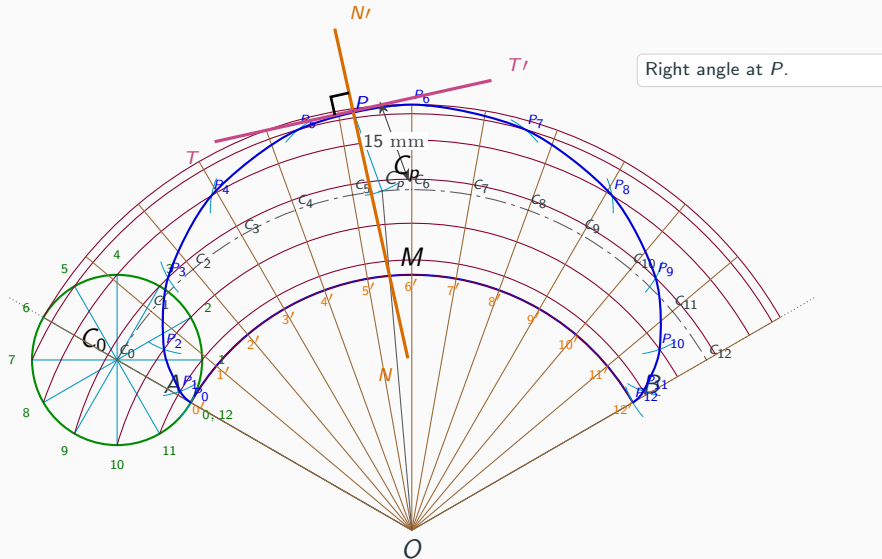
Step 19: Draw the normal NN' through P



Step 20: Draw the tangent perpendicular to NN'



Step 20: Show the right-angle at P



Step 20: Final completed epicycloid construction

